

2020 AMC 10B Problem 17 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 17

Problem:

There are 10 people standing equally spaced around a circle. Each person knows exactly 3 of the other 9 people: the 2 people standing next to her or him, as well as the person directly across the circle. How many ways are there for the 10 people to split up into 5 pairs so that the members of each pair know each other?

Answer Choices:

- A. 11
- B. 12
- C. 13
- D. 14
- E. 15

Solution:

Consider the 10 people to be standing in a circle, where two people opposite each other form a diameter of the circle. Let us use casework on the number of pairs that form a diameter of the circle.

► **Case 1:** 0 diameters

There are 2 ways: either 1 pairs with 2, 3 pairs with 4, and so on, or 10 pairs with 1, 2 pairs with 3, etc.

► **Case 2:** 1 diameter

There are 5 possible diameters to draw (everyone else pairs with the person next to them).

Note that there cannot be 2 diameters since there would be one person on either side that will not have a pair adjacent to them. The only scenario forced is when the two people on either side would be paired up across a diameter. Thus, a contradiction will arise.

► **Case 3:** 3 diameters

There are 5 possible sets of 3 diameters to draw. Notice we are technically choosing the number of ways to choose a pair of two diameters that are neighbors to each other. This means we can choose the first diameter in the pair, and have only two diameters to choose from for the second in the pair. This means we have $5 * 2 = 10$ possibilities for choosing 5 neighboring diameters. However, notice that there are duplicates, so we divide the 10 possibilities by 2 to get 5.

Note that there cannot be a case with 4 diameters because then there would have to be 5 diameters for the two remaining people as they have to be connected with a diameter. A contradiction arises.

► **Case 4:** 5 diameters

There is only 1 way to do this.

Thus, in total there are $2 + 5 + 5 + 1 = (C) \boxed{13}$ possible ways.

OR

If each person knows exactly 3 people, that means we form "4-person groups". That is, all the people in the group know every other person. We can rotate until the circle is filled. We want pairs of people, so the first pair has $\binom{4}{2} = 6$. The 2nd pair is just $\binom{2}{2} = 1$. We need to multiply these together since these are 1 group. The 3rd pair would be $\binom{4}{2} = 6$. The 4th pair is $\binom{2}{2} = 1$. We multiply these 2 together and get 6. The final group would be $\binom{2}{2} = 1$. So we add these up and we have $6 + 6 + 1 = (C) \boxed{13}$ possible ways.

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